

Depth-Dominant Skeleton Detection for Natural Scenes

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Abstract

To date, all natural scene skeleton detection follows the paradigm of taking RGB images as the sole input; despite notable progress, methods under this paradigm suffer significant performance degradation on complex-content images. We observe that depth images are inherently insensitive to color and texture, and can provide clear regional contours and inter-region spatial relationships, which naturally alleviates the difficulty of skeleton detection in complex scenarios. Motivated by this observation, this paper proposes for the first time a novel skeleton detection paradigm where depth images serve as the dominant modality and RGB images act as the auxiliary, and accordingly presents a model DDSkel (short for **Depth-Dominant Skeleton Detection**) under this paradigm. DDSkel employs an asymmetric encoder design to fuse RGB information into depth features, with the RGB modality branch having only 12% the parameters of the depth modality branch. DDSkel has a simple structure without intricate designs. Nevertheless, with only 36% of the trainable parameters of the current best method, DDSkel outperforms all state-of-the-art approaches on SymPASCAL, the most challenging dataset with a large volume of complex images.

Introduction

Generally speaking, a classic skeleton, a key technique for object shape representation, is the medial axis of an object, reduced to a single-pixel-wide line while preserving its topological structure and key shape features [Blum, 1973]. Early skeleton detection methods process binary images only [Bai et al., 2007, Sana et al., 2016]; however, as its application scope expanded, especially in the era of deep learning, more and more algorithms have begun to extract skeletons from natural images [Shen et al., 2017, Zhang et al., 2022, Bai et al., 2023, Wang et al., 2024]. Skeleton detection in natural images has a wide range of applications, such as object

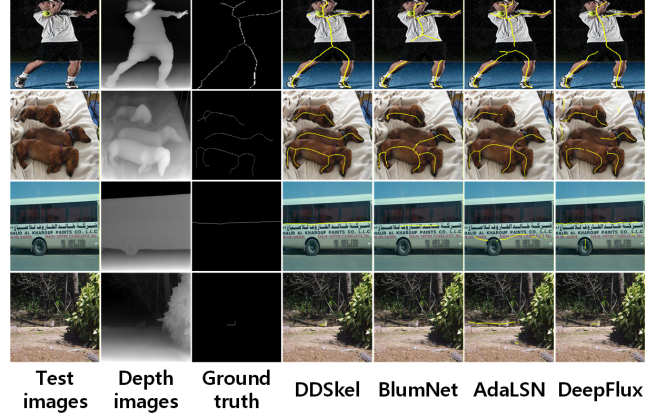


Figure 1: Examples of the comparison of detection results between DDSkel and 3 state-of-the-art skeleton detection methods, BlumNet, AdaLSN, and DeepFlux.

recognition [Li et al., 2024a], text processing [Xie et al., 2024], medical image analysis [Chen et al., 2024], object detection [Gupta et al., 2025], industrial inspection [Guo et al., 2025], image and video generation [Wang et al., 2025], pose estimation [Xu et al., 2025] and remote sensing analysis [Zhou et al., 2025].

All existing skeleton detection methods for natural scenes follow a common paradigm: models take RGB images as their sole input. Despite the remarkable progress achieved in this line of research, methods built on this paradigm still suffer from significant performance degradation when processing complex, cluttered natural scenes. The examples in Fig.1 illustrate this phenomenon. The first column of Fig.1 are the test images; the second column shows the corresponding depth images (generated by DepthAnythingV2 [Yang et al., 2024]), and the third column is the groundtruth; the fourth to the last column display the results of skeleton detection on these images using our proposed DDSkel and other state-of-the-art methods (BlumNet [Zhang et al., 2022], AdaLSN [Liu et al., 2021], and DeepFlux [Wang et al., 2019]). The yellow lines in these results are detected skeletons. In the test images of Fig.1, the

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foreground objects have color or texture regions similar to the background, or have parts occluded, or the content of the foreground objects or background regions is rather complex. These images pose significant challenges for current skeleton detection methods. As can be seen in the results of Fig.1, none of them yield satisfactory detections.

The phenomenon illustrated in Fig.1 is essentially rooted in the inherent nature of the skeleton detection task. The fundamental goal of skeleton detection is to extract the medial axis of object shapes, a task inherently independent of object categories, textures, colors, and background appearance. However, RGB images, especially in complex scenes, inevitably introduce substantial shape-irrelevant interference: complex internal object textures, foreground-background color confusion, texture similarity between targets and their surroundings, and visual ambiguities caused by occlusions. These distractions are the root cause of the drastic performance degradation of existing methods in complex scenarios. To mitigate these issues, existing methods have to resort to designing more complex networks; yet, the limited scale of current natural scene skeleton detection datasets makes such overly complex models highly prone to overfitting.

We notice that depth images are inherently insensitive to color and texture cues, and can provide clear region contours as well as spatial depth relationships across regions, which naturally alleviates the inherent difficulty of skeleton detection in complex scenes. The second column of Fig.1 clearly demonstrates three key advantages of depth images: (1) Depth images contain minimal redundant details in both foreground and background regions, which significantly suppresses the shape-irrelevant interference introduced by cluttered content in RGB images; (2) Depth images capture object contour information more reliably than RGB images. Such contour cues are critical for skeleton detection; and (3) Depth images encode scene geometric depth information, which effectively facilitates skeleton detection in scenes with occlusions and multiple overlapping objects.

In recent years, driven by the rapid development of large models, monocular depth estimation technology has become highly mature [Yang et al., 2024]. State-of-the-art methods in this field are now capable of producing high-quality depth images, such as DepthAnythingV2.

Furthermore, as existing skeleton detection datasets only annotate ground-truth skeletons for specific object categories, while depth images inherently lack category-level semantic information, RGB images are still required to provide auxiliary semantic guidance. In other words, the RGB modality is not entirely redundant.

Based on the above analysis, this paper breaks away from the long-standing paradigm of taking RGB images as the sole input modality, and proposes a novel paradigm for skeleton detection that takes depth images as the dominant modality, with RGB images providing auxiliary semantic guidance. This paradigm fully leverages the complementary strengths of depth and RGB images for the skeleton detection task. Guided by this paradigm, we present DDSkel, a novel skeleton detection model that takes RGB images and depth images as input.

DDSkel follows an encoder-decoder architecture: its encoder adopts an asymmetric dual-branch design that unidirectionally fuses RGB features into depth features, where the lightweight RGB branch contains only 12% of the parameter count of the depth branch. The augmented depth features are then fed into the decoder to predict skeleton. To demonstrate the inherent advantages of the proposed paradigm, we keep the network structure of DDSkel as simple as possible. With only 36% of the trainable parameters of the current best method, DDSkel outperforms all existing state-of-the-art approaches on SymPASCAL, the most challenging dataset with many complex images.

The main contributions of this paper are as follows:

- For the first time, we propose a novel depth-dominant paradigm for skeleton detection, revealing that for skeleton detection in natural scenes, the depth modality, rather than the RGB modality, should preferably dominate the feature encoding process.
- We present DDSkel, a new skeleton detection model built under this paradigm, which takes RGB images and their corresponding depth images as input. With a deliberately designed simple architecture, DDSkel adopts an asymmetric dual-branch fusion encoder structure, achieving efficient skeleton detection via the design of a depth branch paired with a much lighter RGB branch in the encoder.
- Extensive experiments demonstrate that the proposed paradigm effectively overcomes the inherent limitations of RGB-only paradigm. DDSkel delivers significantly better detection results on complex images. Notably, with only 36% of the trainable parameters of the current best method, DDSkel outperforms all existing state-of-the-art approaches on the challenging SymPASCAL dataset.

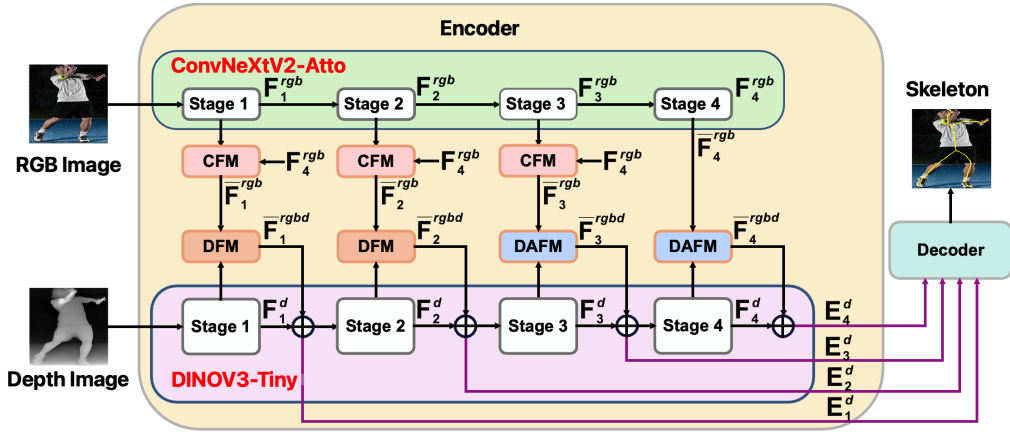


Figure 2: The overall architecture of DDSkel, which consists of an encoder and a decoder. The encoder comprises two backbones, namely ConvNeXtV2-Atto and DINOv3-Tiny, which process RGB images and depth images respectively. RGB features are fused with depth features via the CFM (Cross-stage Fusion Module), DFM (Dual-modality Fusion Module), and DAFM (Dual-modality Attention-based Fusion Module) to enhance depth features, which are then fed into the decoder to produce the final skeleton detection results.

Related Work

CNN-based Skeleton Detection.

The vast majority of existing deep learning methods for skeleton detection are built upon convolutional neural networks (CNNs), centered on manually designed network architectures to boost detection performance. Specifically, Multi-scale Bidirectional Fully Convolutional Network (MSB-FCN) is proposed to better capture and integrate high-level context across multiple scales [Yang et al., 2018, 2021]. A hierarchical feature integration mechanism (Hi-Fi) is introduced to refine multi-level feature aggregation [Zhao et al., 2018]. Fully convolutional networks equipped with scale-associated side outputs are developed to tackle modeling challenges in skeleton prediction [Shen et al., 2017, 2016b]. DeepFlux predicts two-dimensional vector fields via CNN training, mapping each scene pixel to candidate skeleton points for detection [Wang et al., 2019, Xu et al., 2021]. Linear Span Network (LSN) with dedicated linear span units is designed to minimize reconstruction error during skeleton detection [Liu et al., 2018]. ProMask presents a skeleton probability representation that explicitly encodes skeleton pixels through progressive signals [Bai et al., 2023, 2020]. Side-output Residual Network (SRN) leverages error propagation across scales to simplify the fitting of complex outputs with limited network layers [Ke et al., 2017], and Rich Side-output Residual Network (RSRN) further fuses side outputs in a deep-to-shallow hierarchical manner to reduce prediction residuals [Liu et al., 2017a]. Beyond manual architecture design, neural architecture search (NAS) has been applied to automat-

ically discover optimal network structures for skeleton extraction [Qiao et al., 2021, Liu et al., 2021]. Additional distinctive network designs have also been explored to advance skeleton detection performance [Liu et al., 2017b, Fan et al., 2023, Fu et al., 2022, Hou et al., 2018, Xu et al., 2019].

Transformer-based Skeleton Detection.

Departing from the dominant CNN paradigm, transformer architectures have recently been introduced to skeleton detection. Unlike most methods that derive results from predicted skeleton heatmaps, BlumNet directly predicts skeletons via a graph decomposition and reconstruction strategy, with an efficient encoder-decoder scheme optimized under an extended transformer architecture [Zhang et al., 2022]. Additionally, an emerging work revisits the core optimization objective of the skeleton detection task: it proposes to guide the model to gradually shift its focus from coordinate localization regression to point classification during training, and this training paradigm shift can be achieved solely by changing the number of object queries in the Transformer architecture [Wang et al., 2024].

All the aforementioned skeleton detection methods follow the paradigm of taking RGB images as the sole input modality, and suffer from significant performance degradation when processing complex natural scene images. In contrast, the proposed DDSkel adopts our novel depth-dominant, RGB-assisted paradigm, and significantly outperforms all the above methods for skeleton detection in complex scenes.

Methodology

Overview

The overall architecture of the proposed DDSkel is illustrated in Fig.2. Overall, DDSkel follows a standard encoder-decoder structure: the encoder unidirectionally fuses RGB features into depth features, while the decoder leverages the augmented depth features to produce final skeleton predictions. The encoder adopts an asymmetric dual-branch design with two distinct backbones (DINOv3-Tiny [Simeoni et al., 2025] for the depth branch, ConvNeXtV2-Atto [Woo et al., 2023] for the RGB branch) to process the two modalities separately. Notably, the lightweight RGB branch, due to its auxiliary role, contains only 12% of the parameter count of the depth branch. Multi-scale features from the four stages of the RGB branch are injected into the corresponding stages of the depth branch via dedicated cross-modal fusion modules. Finally, the augmented depth features are fed into an FPN-style decoder, which fuses multi-scale representations to output the final skeleton prediction map.

Details of each aforementioned component are elaborated in the following subsections.

Generating Depth Images

Depth images for DDSkel are generated by feeding input RGB images into the monocular depth estimation model DepthAnythingV2 [Yang et al., 2024]. For all training and test images, depth images are generated once and reused across all stages, requiring no redundant computation during either training or inference.

It is worth noting that DepthAnythingV2 is not the only option for our proposed paradigm, but merely a replaceable implementation example. Other state-of-the-art monocular depth estimation methods can also be seamlessly integrated into DDSkel.

Analysis of Depth Images Quality Degradation

Depth images captured by physical sensing devices and those generated by high-performance monocular depth estimation algorithms such as DepthAnythingV2 fundamentally differ in the nature of their quality defects [Yang et al., 2024].

Sensor-captured depth images suffer from inherent acquisition-level flaws, including random pixel noise, large-scale missing holes, over-smoothed object boundaries, and severe failures on transparent surfaces. For skeleton extraction tasks, which rely heavily on the topological integrity of object boundaries, the holes and

noise in sensor depth images can easily lead to fractured, shifted, or even completely missing skeletons.

In contrast, depth images produced by DepthAnythingV2 are free from holes and noise, with well-preserved fine-grained structural details. Its potential errors mainly manifest in two aspects: (1) Under extreme out-of-distribution scenarios, errors present as semantic-level misjudgments of relative depth relations rather than corruption of local geometric structures. (2) In some specific transparent/reflective scenarios, errors appear as deviations in depth values of transparent or reflective regions, while the object contours in the depth image remain intact. In both cases, the edges and geometric shapes of target objects remain continuous and intact, which preserves stable and continuous skeleton topology and still provides reliable shape priors for skeleton detection.

Although depth images generated by DepthAnythingV2 rarely exhibit quality issues on current skeleton detection datasets, we still adopt necessary architectural designs to mitigate this potential problem. Specifically, in DDSkel, we incorporate RGB features into depth features via the CFM, DFM, and DAFM modules to enhance depth representations and alleviate potential depth quality degradation.

The Backbones for Depth and RGB Image

In the encoder of DDSkel, we select the lightweight variant of the recently released DINOv3 [Simeoni et al., 2025], namely DINOv3-Tiny (i.e., the ConvNeXt-based Tiny version of DINOv3), as the backbone for the depth image branch. This choice is motivated by the following considerations. Compared with traditional supervised backbones (e.g., convolutional neural networks and Transformers), DINOv3 yields visual representations with stronger generalization capability and higher robustness through large-scale self-supervised learning. It can not only encode rich semantic information but also accurately preserve the geometric boundaries and structural details of objects [Simeoni et al., 2025], which is highly aligned with the demands of skeleton detection tasks. As the lightweight variant of DINOv3, DINOv3-Tiny inherits all these merits while maintaining a parameter count of only 29M, making it well suited for skeleton extraction, a task with limited dataset scale.

For the RGB image branch, in light of its role as a mere auxiliary branch, we adopt ConvNeXtV2-Atto (with only 3.7M parameters) [Woo et al., 2023], the smallest variant of the ConvNeXtV2 family [Woo et al., 2023], as the backbone. Despite its compact parameter size, ConvNeXtV2-Atto still provides sufficient high-level semantic cues to distinguish target objects from

the background, with negligible extra inference overhead. Furthermore, the shared architectural lineage of ConvNeXtV2-Atto and DINOv3-Tiny facilitates the alignment and fusion of multi-stage features between the RGB and depth branches.

Formally, given the input RGB image I_{rgb} , we use the DepthAnythingV2 model to generate the corresponding depth image I_d . Then, I_{rgb} and I_d are respectively passed through their own backbones to obtain their respective feature sets $\mathbf{F}_{rgb} = \{\mathbf{F}_i^{rgb}\}_{i=1}^4$ and $\mathbf{F}_d = \{\mathbf{F}_i^d\}_{i=1}^4$, where $\mathbf{F}_i^{rgb}$ and $\mathbf{F}_i^d$ are features of each stage of two backbones.

Enhancing Depth Features With RGB Features

As mentioned earlier, while depth images are inherently better suited for skeleton extraction, RGB features are still required as auxiliary cues to augment depth features for the task, given that existing skeleton extraction datasets only annotate skeletons for specific object categories. Additionally, RGB features can also alleviate potential depth image quality degradation through complementary fusion with depth features.

Cross-stage Fusion Module (CFM).

In the encoder of DDSkel, the four-stage RGB features $\{\mathbf{F}_i^{rgb}\}_{i=1}^4$ are not directly applied to enhance the corresponding depth features $\{\mathbf{F}_i^d\}_{i=1}^4$. Instead, $\mathbf{F}_1^{rgb}$, $\mathbf{F}_2^{rgb}$, and $\mathbf{F}_3^{rgb}$ are first fused with $\mathbf{F}_4^{rgb}$ via the CFM module, respectively, before being incorporated to augment $\{\mathbf{F}_i^d\}_{i=1}^4$.

The computation of the CFM module is given by Eq.1:

$$\bar{\mathbf{F}}_{i \in \{1,2,3\}}^{rgb} = CONV_{1 \times 1 \downarrow}(CAT(\mathbf{F}_i^{rgb}, UP(\mathbf{F}_4^{rgb}))), \quad (1)$$

where UP represents an upsampling operation that resizes $\mathbf{F}_4^{rgb}$ to the same spatial resolution as $\mathbf{F}_{i \in \{1,2,3\}}^{rgb}$; CAT denotes concatenation along the channel dimension, and $CONV_{1 \times 1 \downarrow}$ represents the 1×1 convolution that reduces the channel dimension back to the original dimension of $\mathbf{F}_i^{rgb}$; $\bar{\mathbf{F}}_4^{rgb}$ is simply $\mathbf{F}_4^{rgb}$ itself.

The design of CFM is motivated by two considerations: (1) $\mathbf{F}_4^{rgb}$ contains the most abundant RGB-derived high-level semantic information, which should participate in the fusion of depth features across all 4 stages; (2) $\mathbf{F}_1^{rgb}$, $\mathbf{F}_2^{rgb}$, and $\mathbf{F}_3^{rgb}$ provide multi-scale RGB-related semantic cues, which can guide the optimization of depth features $\{\mathbf{F}_i^d\}_{i=1}^4$ toward skeleton extraction at different granularities.

Dual-modality Attention-based Fusion Module (DAFM).

Features $\bar{\mathbf{F}}_3^{rgb}$ and $\bar{\mathbf{F}}_4^{rgb}$ are used to fuse and augment depth features $\mathbf{F}_3^d$ and $\mathbf{F}_4^d$ via the DAFM, which comprises two sequentially connected components: a spatial attention module followed by a channel attention module.

Specifically, the spatial attention module is based on the AttnZero structure [Li et al., 2024b], which is the first framework designed to automatically discover efficient attention modules tailored for vision Transformers. As a high-quality linear attention variant, it strikes a favorable balance between linear computational complexity and cross-model generalizability, enabling effective global spatial attention at a low computational cost.

Prior to the AttnZero computation, $\bar{\mathbf{F}}_3^{rgb}$ and $\bar{\mathbf{F}}_4^{rgb}$ are channel-wise concatenated with $\mathbf{F}_3^d$ and $\mathbf{F}_4^d$, respectively, yielding feature $\mathbf{F}_3^{rgbd}$ and $\mathbf{F}_4^{rgbd}$. Each of them is then linearly projected into query (Q), key (K), and value (V), before being fed into the AttnZero module. The detailed computation of AttnZero is formulated as follows (Li et al. 2024b):

$$AttnZero = Q \times \varphi_2 \left(\varphi_2(Q) \varphi_2 \left(\beta_1(K)^T \varphi_1(V) \right) \right) \quad (2)$$

where β_1 is the Exponential Linear Unit (ELU) with a shift of 1; φ_1 denotes min-max normalization; φ_2 is l2 normalization function.

For the channel attention in DAFM, we adopt the Channel Attention Module (CAM) from the CBAM model [Woo et al., 2018]. It is worth noting that since the focus of this work is to present a novel depth-dominated paradigm for skeleton extraction rather than to design sophisticated network architectures, we directly adopt simple and well-established existing modules to construct DAFM.

For illustration, we take $\mathbf{F}_4^{rgbd}$ as a representative of $\mathbf{F}_3^{rgbd}$ and $\mathbf{F}_4^{rgbd}$, and detail its computation flow in the DAFM module as follows:

$$\bar{\mathbf{F}}_4^{rgbd} = CONV_{1 \times 1 \downarrow}(CAM(AttnZero(\mathbf{F}_4^{rgbd}))) \quad (3)$$

where $CONV_{1 \times 1 \downarrow}$ denotes the operation that reduces the channel dimension of the output feature to match that of $\mathbf{F}_4^d$. Feature $\bar{\mathbf{F}}_3^{rgbd}$ is computed in the same manner.

The design motivations of DAFM are twofold. (1) Features $\bar{\mathbf{F}}_3^{rgb}$ and $\bar{\mathbf{F}}_4^{rgb}$, which carry richer high-level semantic information, are critical for depth feature enhancement. Accordingly, we implement the augmentation of depth features $\mathbf{F}_3^d$ and $\mathbf{F}_4^d$ through dual spatial and channel attention that captures global contextual

information. (2) When depth images suffer from quality degradation, DAFM can suppress the involvement of degraded depth features in the computation by adaptively weighting the importance of RGB features and depth features.

Dual-modality Fusion Module (DFM).

RGB features $\bar{\mathbf{F}}_1^{rgb}$, $\bar{\mathbf{F}}_2^{rgb}$ and depth features $\mathbf{F}_1^d$, $\mathbf{F}_2^d$ have higher spatial resolution. To reduce computational overhead, we adopt a simpler DFM structure for the fusion of these features. Similar to DAFM, before the computation of DFM, $\bar{\mathbf{F}}_1^{rgb}$ and $\bar{\mathbf{F}}_2^{rgb}$ are channel-wise concatenated with $\mathbf{F}_1^d$ and $\mathbf{F}_2^d$, respectively, yielding feature $\mathbf{F}_1^{rgb d}$ and $\mathbf{F}_2^{rgb d}$. Then, taking feature $\mathbf{F}_1^{rgb d}$ as an example, the computation of DFM is formulated as follows:

$$\bar{\mathbf{F}}_1^{rgb d} = CONV_{1 \times 1 \downarrow}(\mathbf{F}_1^{rgb d}) \quad (4)$$

where the function of $CONV_{1 \times 1 \downarrow}$ is similar to the one in Eq.3. Feature $\bar{\mathbf{F}}_2^{rgb d}$ is computed in the same way.

The above-computed features $\bar{\mathbf{F}}_1^{rgb d}$, $\bar{\mathbf{F}}_2^{rgb d}$, $\bar{\mathbf{F}}_3^{rgb d}$, and $\bar{\mathbf{F}}_4^{rgb d}$ are directly added element-wise with their corresponding depth features $\mathbf{F}_1^d$, $\mathbf{F}_2^d$, $\mathbf{F}_3^d$, and $\mathbf{F}_4^d$, respectively, yielding enhanced depth features $\mathbf{E}_1^d$, $\mathbf{E}_2^d$, $\mathbf{E}_3^d$, and $\mathbf{E}_4^d$ that serve as inputs to the decoder.

Decoder

The decoder is constructed in an FPN-like manner [Lin et al., 2017]. Features $\mathbf{E}_1^d$, $\mathbf{E}_2^d$, $\mathbf{E}_3^d$, and $\mathbf{E}_4^d$ first undergo channel dimension reduction via 1×1 convolutions. They are then sequentially processed through upsampling, element-wise addition with corresponding higher-resolution features, and ConvBNSiLU units. Each ConvBNSiLU unit consists of a 3×3 convolution, a Batch Normalization (BN) layer, and a SiLU [Hendrycks and Gimpel, 2016] activation function. The final output of the decoder is the skeleton detection map. The detailed architecture of the decoder is illustrated in the supplementary material.

Loss Function

Let $\mathbf{P} \in [0, 1]$ denotes the predicted skeleton probability map and $\mathbf{G} \in [0, 1]$ as the binary groundtruth. When training DDSkel, the loss function is a combination of a weighted ℓ_2 loss [Bai et al., 2020] and a soft Dice loss [Milletari et al., 2016].

Weighted ℓ_2 loss. To mitigate the imbalance in the number of skeleton and non-skeleton pixels, the

weighted ℓ_2 loss [Bai et al., 2020] is defined as:

$$\mathcal{L}_{w\ell_2} = \sum_{i=1}^N \alpha_i (\mathbf{G}_i - \mathbf{P}_i)^2, \quad (5)$$

where i indexes pixels and N is the number of pixels in a mini-batch. The balance weight α_i is inversely proportional to the ratio of the number of 0 and 1 pixels.

Soft Dice loss. We also enforce region-level agreement using the soft Dice loss [Milletari et al., 2016]:

$$\mathcal{L}_{sDice} = 1 - \frac{2 \sum_{i=1}^N \mathbf{P}_i \mathbf{G}_i + \epsilon}{\sum_{i=1}^N \mathbf{P}_i + \sum_{i=1}^N \mathbf{G}_i + \epsilon}, \quad (6)$$

where ϵ is a small constant for numerical stability.

The total loss is:

$$\mathcal{L} = \mathcal{L}_{w\ell_2} + 0.5 \mathcal{L}_{sDice}. \quad (7)$$

Experiments

Datasets and Evaluation Protocol

We conduct experiments on object skeleton detection using four widely-used datasets: **SK-LARGE** [Shen et al., 2017], **SK506** [Shen et al., 2016b], **WH-SYMMAX** [Shen et al., 2016a], and **SymPASCAL** [Ke et al., 2017]. Among them, the SymPASCAL dataset contains a large volume of complex uncropped in-the-wild images with cluttered backgrounds and dense distracting textures, making it highly challenging and significantly increasing the difficulty of skeleton detection. The datasets splits are as follows: SK506 contains 300 training images and 206 testing images; WH-SYMMAX contains 328 images, where the first 228 images are used for training and the remaining 100 images are used for testing; SK-LARGE contains 746 training images and 745 testing images; SymPASCAL contains 648 training images and 787 testing images under 20 object classes.

Like other state-of-the-art methods, in our experiments, we use the F-measure metric [Wang et al., 2019] to evaluate the performance of skeleton detection.

Implementation Details

Details of Training.

The training batch size is 4, while the validation and test batch size is 2. All images are resized to 512×512 . All experiments use multi-GPU distributed training on NVIDIA RTX 4090 GPUs. We optimize the network using AdamW with module-specific parameter groups.

The default group, including the backbones and decoder, uses a learning rate of 1×10^{-5} , while the fusion-related modules use 5×10^{-4} . The weight decay is 1×10^{-4} .

We employ CosineAnnealingLR with a linear warmup. During the first 5 epochs, the learning rate increases linearly from 1×10^{-6} to the target learning rate of each parameter group, and then follows cosine decay to 1×10^{-7} . The model is trained for 65 epochs. Data augmentation applies synchronized geometric transformations to the RGB image, ground-truth skeleton, and depth image, including random horizontal flipping, random vertical flipping, and a random rotation sampled from $\{90^\circ, 180^\circ, 270^\circ\}$.

Details of Encoder and Decoder.

The depth primary backbone is DINOv3-Tiny, which contains four stages with channel dimensions [96, 192, 384, 768] and block depths [3, 3, 9, 3]. The RGB auxiliary backbone is ConvNeXtV2-Atto. Its four stages use channel dimensions [40, 80, 160, 320] and block depths [2, 2, 6, 2].

For a 512×512 input, the four-stage feature-map resolutions are 128×128 , 64×64 , 32×32 , and 16×16 . During training, all parameters of both the encoder and decoder are trainable. The decoder adopts an FPN-like structure with a hidden dimension of 256 and SiLU activation.

Results

Overall Quantitative Comparison.

Tab. 1 presents a quantitative comparison between our DDSkel and 9 state-of-the-art methods (HED [Xie and Tu, 2015], SRN [Ke et al., 2017], Hi-Fi [Zhao et al., 2018], DeepFlux [Wang et al., 2019], AdaLSN [Liu et al., 2021], FMRN [Fan et al., 2023], ProMask [Bai et al., 2023], BlumNet [Zhang et al., 2022], BlumNet+ [Wang et al., 2024]) for skeleton detection in natural images on 4 datasets, evaluated by F-measure scores. Values highlighted in bold in Tab. 1 correspond to the highest-ranked F-measure scores. As can be observed from the table, on SympASCAL, which contains a large number of challenging images, DDSkel achieves significantly superior performance over all competing methods. Its F-measure outperforms the second-best method, BlumNet+, by 10%, yet DDSkel only accounts for 36% of the parameters of BlumNet+, which can be seen from Tab. 2. This verifies that DDSkel yields notably better performance with far fewer parameters on complex images.

Furthermore, on the remaining three datasets with much fewer complex images, DDSkel either outperforms

Method	SK506	SK-LARGE	WH-SYMMAX	SymPASCAL
HED	0.542	0.497	0.732	0.369
SRN	0.632	0.658	0.780	0.443
Hi-Fi	0.681	0.724	0.805	0.454
DeepFlux	0.695	0.732	0.840	0.502
AdaLSN	0.740	0.786	0.851	0.497
FMRN	0.751	0.789	0.865	0.508
ProMask	0.754	0.772	0.875	0.582
BlumNet	0.752	0.826	0.877	0.521
BlumNet+	0.799	0.836	0.913	0.612
DDSkel	0.780	0.825	0.924	0.674

Table 1: Quantitative comparison on 4 datasets with 9 methods using F-measure scores. Values highlighted in bold in table correspond to the highest-ranked F-measure scores.

Method	Params (M)	FLOPs (G)	Time (ms)
HED	14.72	103.75	5.18
SRN	14.72	103.75	5.23
DeepFlux	18.32	125.93	6.39
Ada-LSN	32.38	217.07	15.05
BlumNet	99.32	121.0	22.8
BlumNet+	100.80	—	—
DDSkel	35.66	41.94	12.32

Table 2: Computational-efficiency comparison with existing skeleton-detection methods. The symbol ‘—’ means no results provided by authors.

or is highly comparable to the best-performing methods. Specifically, on WH-SYMMAX, DDSkel leads the second-best BlumNet+ by 1%, while on SK506 and SK-LARGE, its performance is highly close to that of the top-ranked method. Notably, across all 4 datasets, DDSkel consistently and significantly outperforms the methods with a similar parameter count (see Tab. 2).

Tab. 2 presents a comparison of computational efficiency between DDSkel and 6 state-of-the-art methods, covering parameter count, FLOPs, and per-image inference time. These methods are selected as their published papers or publicly available source codes allow us to obtain part or all of these metrics. Combining the results from Tab. 1 and Tab. 2, it can be observed that DDSkel achieves the best detection performance among methods with comparable parameter scales.

Overall Qualitative Comparison.

Fig.3 shows some qualitative comparison between DDSkel and 3 state-of-the-art methods (BlumNet, AdaLSN and DeepFlux). The test images in Fig.3 all have complex content. However, in the corresponding depth images, these complex interferences are effectively suppressed. As can be seen, the skeleton detected by DDSkel is the closest to the groundtruth, while the other comparison methods have obvious detection errors. More qualitative comparisons between DDSkel

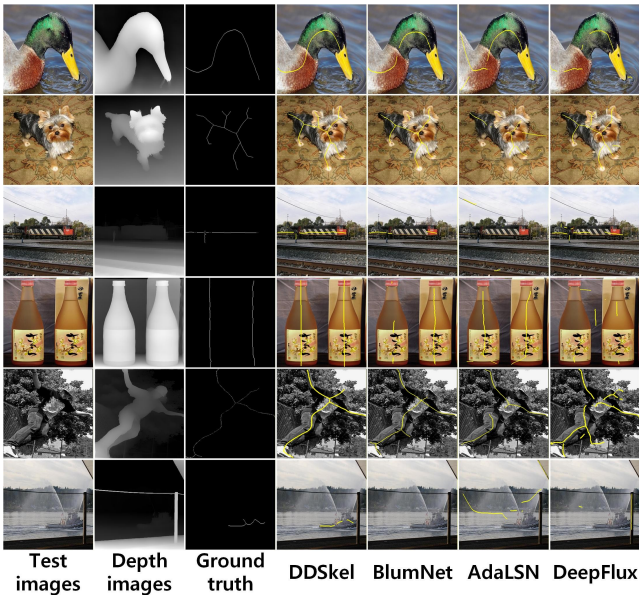


Figure 3: The qualitative comparison between DDSkel and three state-of-the-art methods, BlumNet, AdaLSN, and DeepFlux.

Configuration	SymPASCAL	SK-LARGE
RGB-Only	0.6197	0.8008
Depth-Only	0.6602	0.8051
RGB-RGB	0.6345	0.8046
Depth-Depth	0.6633	0.8102
RGB-Depth	0.6622	0.8138
Depth-RGB(DDSkel)	0.6740	0.8250

Table 3: Comparison of different modality assignments under a fixed network architecture and training protocol. For dual-branch settings, the first modality is assigned to the primary backbone and the second modality to the auxiliary branch.

and state-of-the-art methods are provided in the supplemental material of this paper.

Ablation Study

Ablation on Dual-backbone and Depth Primacy.

This ablation study justifies the dual-backbone design of the encoder with both RGB and depth modalities in DDSkel, as well as the rationale for adopting depth as the primary modality. Tab. 3 presents the results of this ablation study.

In the first column of Tab. 3, each variant is defined as follows. RGB-Only: Only the larger backbone in the encoder takes RGB images as input, while the smaller backbone is removed. Depth-Only: Only the larger backbone takes depth images as input, with the

smaller backbone removed. RGB-RGB: Both backbones of the encoder receive RGB images as input. Depth-Depth: Both backbones of the encoder are fed with depth images. RGB-Depth: The larger backbone takes RGB images as input, while the smaller backbone takes depth images as input. The second and third columns of Tab. 3 report the F-measure values evaluated on the two datasets, respectively.

We take the results on SymPASCAL from Tab. 3 as an example for analysis. For the single-backbone baselines, Depth-Only clearly outperforms RGB-Only on SymPASCAL (0.6602 to 0.6197), indicating that geometric structure provides a more suitable primary representation for skeleton extraction. Although dual-backbone models introduce additional parameters, their gains cannot be attributed merely to increased model capacity. Depth-Depth improves only slightly over Depth-Only on SymPASCAL (0.6602 to 0.6633), showing that simply duplicating the depth branch offers limited benefit. In contrast, introducing a complementary RGB auxiliary branch raises the Depth-RGB result to 0.6740, outperforming both Depth-Depth and the role-reversed RGB-Depth configuration (0.6622). These results indicate that the gain mainly arises from the complementary roles of Depth and RGB rather than a simple increase in dual-backbone capacity.

Ablation on RGB Auxiliary Backbone.

In this ablation study, we replace the RGB backbone in the encoder with larger ConvNeXtV2 variants. Tab. 4 shows the corresponding results on SymPASCAL and SK-LARGE. We fix the depth branch and the fusion configuration, and vary only the RGB auxiliary backbone. The 3 replacement RGB backbones are ConvNeXtV2-Tiny, ConvNeXtV2-Pico, and ConvNeXtV2-Femto. The results in Tab. 4 demonstrate that adopting a larger RGB backbone yields only marginal detection improvements but significantly increases the parameter count, making it an unfavorable trade-off. We ultimately select ConvNeXtV2-Atto as the default configuration, as it strikes a favorable balance between performance and efficiency.

Other Ablation Studies.

We also conduct ablation studies on CFM, DFM, DAFM module, and how these modules help DDSkel address potential quality degradation issues in depth images. Due to space constraints, we defer the presentation and discussion of these ablation results to the supplementary material.

RGB Auxiliary Branch	SymPASCAL SK-LARGE	
ConvNeXtV2-Tiny (60.85M)	0.6802	0.8292
ConvNeXtV2-Pico (41.15M)	0.6753	0.8264
ConvNeXtV2-Femto (37.23M)	0.6759	0.8247
ConvNeXtV2-Atto (35.66M, Ours)	0.6740	0.8250

Table 4: Ablation of the RGB auxiliary-branch capacity with the depth primary branch and fusion configuration fixed. The values in parentheses denote the total parameter count of the complete dual-stream model. Values highlighted in bold in table correspond to the highest-ranked F-measure scores.

Conclusion

This paper re-examines the mainstream RGB-only paradigm for natural scene skeleton detection, and points out that such a paradigm suffers from severe performance degradation when processing complex-content images. To tackle this limitation, we propose for the first time a novel depth-dominant skeleton detection paradigm, arguing that depth modality is more suitable to dominate the feature encoding process for skeleton detection tasks. Under this paradigm, we build DDSkel, an efficient skeleton detection model with an asymmetric dual-branch encoder architecture. Extensive experiments verify that the depth-dominant paradigm can effectively overcome the inherent limitations of the RGB-only paradigm. For future work, we will further explore the potential of depth information in skeleton extraction and design more lightweight and efficient models.

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